



Carisurg

Assignment 6

Triage Pseudocode

Due Date: May 26th, 2025.

Prepared by **Josiah-John Green**

Digital Triage System — ESI Level Classifier

This system replicates the ESI (Emergency Severity Index) 5-level triage framework used at Mercer General ED. It takes patient vitals and demographic data captured at registration and triage, and returns a recommended ESI level along with a clinical justification.

Important: This is decision SUPPORT, not decision REPLACEMENT.

- A trained triage clinician must always review the output.
- A digital system cannot account for patient affect, appearance, or history the way a nurse at the bedside can.
- Input data available at triage (from the Mercer triage form):
 - Age, Gender
 - Chief complaint
 - GCS (Glasgow Coma Scale)
 - SBP, DBP, MAP (blood pressure)
 - Pulse (heart rate)
 - RR (respiratory rate)
 - Temp (body temperature)
 - SpO2 (oxygen saturation — recommended addition)
 - FiO2 (fraction of inspired oxygen)
 - Pain score (0–10)

Pseudocode:

```
def digital_triage(patient):
    """
    Assess a patient's triage priority using ESI levels 1–5.

    Parameters
    -----
    patient: dict
        Dictionary of patient vitals and demographics collected at triage.

    Returns
    -----
    dict
        ESI level (1–5), label, and clinical justification string.
    """

    # — STEP 1: EXTRACT PATIENT DATA —————
    # Pull each value from the patient record.
    # In a real system, these would come from a form submission or EHR feed.

    age      = patient.get('age')
    gcs       = patient.get('gcs')           # 3–15. Below 8 = severe impairment
    sbp       = patient.get('sbp')           # mmHg. Below 90 = hypotension
    map_val   = patient.get('map')           # mmHg. Below 65 = organ hypoperfusion
    pulse     = patient.get('pulse')         # bpm. Normal: 60–100
    rr        = patient.get('rr')            # breaths/min. Normal: 12–20
    temp      = patient.get('temp')          # °C. Fever: >38.0
    spo2      = patient.get('spo2')          # %. Below 90 = hypoxaemia
```

```

fio2      = patient.get('fio2')          # %. Room air = 21%
pain      = patient.get('pain_score')    # 0-10 self-reported
complaint = patient.get('chief_complaint', '').lower()

# — STEP 2: ESI 1 – IMMEDIATE, LIFE-THREATENING —————
# Any one of these conditions alone warrants ESI 1.
# These patients go directly to the resuscitation bay.
# The most senior clinician on shift takes ownership immediately.

if (
    gcs is not None and gcs <= 8          # severe neurological impairment
    or map_val is not None and map_val < 65 # organ hypoperfusion / shock
    or sbp is not None and sbp < 70       # shock-range hypotension
    or spo2 is not None and spo2 < 85     # severe hypoxaemia
    or rr is not None and (rr < 8 or rr > 35) # respiratory failure range
    or any(word in complaint for word in [
        'cardiac arrest', 'not breathing', 'unconscious',
        'major trauma', 'anaphylaxis'
    ])
):
    return {
        'esi': 1,
        'label': 'Resuscitation – Immediate',
        'justification': (
            'One or more life-threatening parameters detected. '
            'Patient requires immediate intervention. '
            'Direct to resuscitation bay.'
        )
    }

```

```

    }

# — STEP 3: ESI 2 – EMERGENCY, HIGH RISK —————
# Patient is not immediately dying but is at high risk of deterioration.
# Must be seen within 10 minutes. Acute cubicle.

if (
  gcs is not None and gcs <= 13          # altered consciousness
  or map_val is not None and map_val < 70 # approaching hypoperfusion
  or sbp is not None and sbp < 90        # hypotension
  or pulse is not None and pulse > 120   # significant tachycardia
  or pulse is not None and pulse < 50    # significant bradycardia
  or spo2 is not None and spo2 < 92      # hypoxaemia
  or temp is not None and temp > 39.5    # high fever
  or temp is not None and temp < 35.5    # hypothermia
  or pain is not None and pain >= 8      # severe pain
  or any(word in complaint for word in [
    'chest pain', 'stroke', 'seizure', 'overdose',
    'difficulty breathing', 'severe bleeding'
  ])
):
  return {
    'esi': 2,
    'label': 'Emergency – Seen within 10 minutes',
    'justification': (
      'High-risk presentation detected. Vitals or complaint indicate '
      'potential for rapid deterioration. Acute cubicle. '
      'Physician to assess within 10 minutes.'
    )
  }

```

```

    )
}

# — STEP 4: ESI 3 – URGENT, MULTIPLE RESOURCES NEEDED —————
# Vitals not in danger zone but patient needs investigation.
# Labs, imaging, or IV access likely required. Seen within 30 minutes.

if (
  pulse is not None and (pulse > 100 or pulse < 60) # mild abnormality
  or rr is not None and (rr > 20 or rr < 12)        # mild abnormality
  or temp is not None and temp > 38.0              # fever
  or spo2 is not None and spo2 < 95                # mild hypoxaemia
  or pain is not None and pain >= 5                # moderate pain
  or any(word in complaint for word in [
    'abdominal pain', 'vomiting', 'dizziness',
    'confusion', 'infection', 'fall'
  ])
):
  return {
    'esi': 3,
    'label': 'Urgent – Seen within 30 minutes',
    'justification': (
      'Vitals show mild abnormality or complaint suggests multiple '
      'resources needed. Patient stable but requires investigation. '
      'Acute cubicle or observation bay.'
    )
  }

```

```

# — STEP 5: ESI 4 – SEMI-URGENT, ONE RESOURCE NEEDED _____
# Vitals normal. Simple presentation requiring one intervention.
# Fast-track area. Seen within 60 minutes.

if (
  any(word in complaint for word in [
    'laceration', 'sprain', 'urinary tract', 'rash',
    'ear pain', 'eye irritation', 'minor injury'
  ])
  or pain is not None and pain <= 4    # mild pain, stable vitals
):
  return {
    'esi': 4,
    'label': 'Semi-Urgent – Seen within 60 minutes',
    'justification': (
      'Vitals within normal range. Single resource likely needed. '
      'Appropriate for fast-track area.'
    )
  }

# — STEP 6: ESI 5 – NON-URGENT _____
# No abnormal vitals. Minimal or no clinical resources needed.
# Can wait. Fast-track or waiting room.

return {
  'esi': 5,
  'label': 'Non-Urgent – Can wait',

```

```
    'justification': (  
        'No abnormal vitals detected. Presentation does not require '  
        'immediate clinical resources. Fast-track or waiting room.'  
    )  
}
```

```
# — STEP 7: RE-TRIAGE REMINDER —————  
# A digital system must support re-triage, not just initial assessment.  
# Patients triaged ESI 4 or 5 who remain in the waiting room must be  
# re-evaluated at regular intervals. Any deterioration in vitals or  
# new complaints should trigger a re-run of this function and escalation  
# if the ESI level changes.  
#  
# A patient who walked in as ESI 4 can become ESI 2 within the hour.  
# The system must flag this – it cannot assume the first assessment holds.
```

```
# — EXAMPLE USAGE —————  
  
patient_example = {  
    'age': 67,  
    'gcs': 14,  
    'sbp': 88,  
    'dbp': 52,  
    'map': 64,  
    'pulse': 118,  
    'rr': 22,  
    'temp': 38.8,
```



```
'spo2': 93,  
'fio2': 21,  
'pain_score': 6,  
'chief_complaint': 'confusion and difficulty breathing'  
}
```

```
result = digital_triage(patient_example)  
print(f"ESI Level   : {result['esi']}")  
print(f"Label       : {result['label']}")  
print(f"Reasoning    : {result['justification']}")
```